

Dissertation Outline

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1 Working Title

- Proposed dissertation title: EMDA and CFD in Numerical Relativity: (something wittier in the works)

2 Abstract Sketch

2.1 Problem

Introduction of the EMDA extension of binary black hole mergers that we will study. Discussion of how these systems may be distinguishable from GR. Explanation of compact finite difference (CFD) methods and why they are of interest.

2.2 Method

Explanation of the numerical relativity code. Description of initial data generation and evolution methods.

2.3 Results

Comparison of EMDA and GR waveforms. Evaluation of CFD operators compared to explicit finite difference operators.

2.4 Significance

Discussion of EMDA detectability and implications of CFD performance for numerical relativity.

3 Chapter-by-Chapter Outline

3.1 Chapter 1: Introduction and Background

- Introduction
- Background
- Summary of the dissertation

3.2 Chapter 2: EMDA Stability and Initial Data

- Introduction and prior work
- Action and equations
- 3+1 decomposition and BSSN formalism
- Initial data construction
- Results (head-on waveforms)

3.3 Chapter 3: EMDA Inspiring Binary Black Hole Mergers

- Introduction
- Initial data
- Eccentricity reduction discussion
- Description of parameter space
- Waveform mismatch with GR (detectability)

3.4 Chapter 4: Interpolated EMDA Black Holes

- Introduction
- Description of interpolated parameter space
- Initial data
- Effects of coupling parameters
- Mismatch and detectability analysis

3.5 Chapter 5: Compact Finite Difference Operators

- Explanation of CFD methods
- Construction of operators
- Electromagnetic test results
- BSSN test results
- Conclusions

4 Conclusion

- Summary of scientific questions addressed
- Key results
- Interpretation
- Limitations
- Future work

A Mathematical Foundations

A.1 Introduction to Numerical Relativity

A.2 3+1 Decomposition and BSSN/EMDA Derivations

B EMDA Black Hole Solutions

B.1 Analytic EMDA Black Holes

B.2 Properties and Limits

C Waveform Analysis

C.1 Mismatch Calculations

C.2 Waveform Catalog

D Code and Implementation Details

D.1 Dendro Framework Notes

D.2 Algorithmic Details

E Compact Finite Difference Operators

E.1 Derivations

E.2 Operator Tables

F Software and Usage

F.1 Running EMDA Simulations

F.2 Using the CFD operators

F.3 Visualization (ParaView, etc.)

G Timeline

TBD